Weiyan Xie (Vayne)

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Professional Summary

AI researcher and builder focused on translating frontier multimodal methods into practical solutions. I enjoy turning ambiguous problems into structured plans, aligning technical and non-technical stakeholders, and delivering explainable, robust, and efficient systems for real-world use.

Core Strengths

- Independent problem solving from zero to publication: in most Ph.D. projects, I initiated the idea, designed the approach, ran experiments, and drove the work to a publishable outcome.
- Research execution with coordination: while owning the core pipeline end-to-end, I coordinated efficiently with co-authors to align on framing, evidence quality, and final delivery.
- Fast learner in new domains: I quickly absorb unfamiliar methods, tools, and application contexts, then convert them into practical plans and results.
- Clear communication with technical and non-technical audiences: I explain trade-offs in plain language and use structured narratives to support decisions.
- Leadership under ambiguity: led multi-team volunteer initiatives with measurable social impact and sustained execution.

Education

- Ph.D. in Computer Science and Engineering, Hong Kong University of Science and Technology, 2022.09-2026.10 (expected).
- M.Sc. in Big Data Technology, Hong Kong University of Science and Technology, 2019.09-2020.12.
- B.S. in Statistics, Beijing Normal-Hong Kong Baptist University, 2015.09-2019.06.

Real-World AI Impact

- **Effective MLLM reasoning for long documents.** Focus: Bridge the gap between static benchmark-style pipelines and real document workflows where evidence is sparse, distributed, and high-resolution. Contribution: Framed a deployable strategy for iterative evidence gathering so systems can reason more reliably on long, visually complex documents with better efficiency.
- **Controllable image editing for practical workflows.** Focus: Turn image editing models into usable tools for creators who need precise structure control without expensive retraining cycles. Contribution: Designed a practical control workflow that improves edit predictability and user intent alignment while keeping deployment cost low.
- **Explainability for trustworthy model decisions.** Focus: Diagnose what visual evidence models truly rely on and reduce spurious, non-causal behavior. Contribution: Developed interpretable analysis workflows to make model reasoning auditable and actionable for real-world validation and debugging.
- **Generalization and robustness in real-world shift.** Focus: Improve model stability after fine-tuning and under distribution shift. Contribution: Built training/evaluation strategies that improve reliability when models move from curated settings to noisy, shifted real data.

Selected Publications (Condensed)

- InSight-doc: Agentic Visual Perception for Long-Document Understanding. In submission, 2026. * Equal contribution (alphabetical order).
- CannyEdit: Selective Canny Control and Dual-Prompt Guidance for Training-Free Image Editing. IEEE CAI, 2026.
- Dual Risk Minimization: Towards Next-Level Robustness in Fine-Tuning Zero-Shot Models. NeurIPS, 2024. * Equal contribution (alphabetical order).
- Consistency Regularization for Domain Generalization with Logit Attribution Matching. UAI, 2024. * Equal contribution (alphabetical order).
- ViT-CX: Causal Explanation of Vision Transformers. IJCAI, 2023.

Experience

2025-2026 Research Intern, Huawei Hong Kong AI Framework & Data Technologies Lab

- Developed and evaluated practical MLLM/vision solutions with emphasis on deployment constraints, inference efficiency, and reliability.
- Collaborated with researchers and engineers to translate research ideas into reproducible prototypes and measurable improvements.

2018-2019 Vice President, United Innovation Charity Club (UICC), BNBU (largest student club on campus)

- Oversaw 7 charity project groups across migrant-worker children support, environmental protection, and traditional village culture preservation.
- Coordinated approximately 300 volunteers and improved cross-team planning and communication cadence.

2017-2019 Project Leader, Working Group on Primary Student Financial Assistance, United Innovation Charity Club (UICC), BNBU

- Led outreach and fundraising for children of migrant-worker families with financial hardship.
- Raised RMB 40,000+ over two years and supported tuition and living expenses for 24 students.
- Coordinated volunteers, partner communication, and transparent fund allocation.

Teaching & Mentoring

- Teaching Assistant, Postgraduate Machine Learning (MSBD5012 / CSIT5910 / COMP5212): 2022 Fall, 2023 Spring, 2023 Fall, 2024 Fall, 2025 Fall.
- Teaching Assistant, COMP2011 Programming with C++: 2024 Spring.
- Co-supervised 15+ Master's independent projects; most students received A-level grades.

Professional Service

- Conference reviewer / PC member: NeurIPS, ICML, ICLR, UAI, AAAI.

Awards

- Huawei PhD Fellowship (HKUST), 2022-2026.
- MSc Big Data Technology Top Students Award (HKUST), 2020.
- School of Engineering Excellent Student Scholarship (HKUST), 2020.

Technical & Collaboration Skills

- Applied AI: multimodal LLMs, vision-language systems, model evaluation, inference optimization.
- Methods: explainability, robustness, domain generalization, controllable generation/editing.
- Engineering: Python, PyTorch, Hugging Face, vLLM, Docker, Git.
- Collaboration: stakeholder communication, project leadership, mentoring.